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INFORMATION OUTPUT DEVICE, CAMERA, INFORMATION OUTPUT SYSTEM, INFORMATION OUTPUT METHOD, AND PROGRAM

Abstract

An information output device includes: a first output unit that outputs acquired information acquired by a sensor; and a second output unit that converts personal information included in the acquired information into attribute information from which identification of an individual is impossible, and outputs the attribute information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation application of U.S. patent application Ser. No. 18/227,659 filed on Jul. 28, 2023, which is a continuation application of U.S. patent application Ser. No. 17/542,688 filed on Dec. 6, 2021, which issued as U.S. Pat. No. 12,177,191, which is a continuation application of U.S. patent application Ser. No. 16/434,038 filed on Jun. 6, 2019, which issued as U.S. Pat. No. 11,227,061, which is a continuation application of U.S. patent application Ser. No. 15/541,179 filed on Jun. 30, 2017, which is a National Stage Entry of international application PCT/JP2016/051147, filed Jan. 15, 2016, which claims the benefit of priority from Japanese Patent Application 2015-005538 filed on Jan. 15, 2015, the disclosures of all of which are incorporated in their entirety by reference herein.

TECHNICAL FIELD

[0002] The present invention relates to an information output device, a camera, an information output system, an information output method, and a program.

BACKGROUND ART

[0003] The spread of technologies related to IoT (Internet of Things) has led to various solutions being provided which utilize real-world data captured by cameras. As a representative example, security companies remotely monitor image data captured by security cameras and use it for security purposes such as helping crime prevention. Furthermore, a usage method can be considered where an observation camera is installed in a sales area to capture the purchasing behavior of customers, analyze the captured image data, and use it for marketing.

[0004] An image captured by a camera displays a face of an individual and the like, and the handling of the image data thereof presents a problem. Furthermore, although personal information (image information by which an individual can be identified) is necessary in a security-purpose system, in a marketing-purpose system it is sufficient if attribute information such as the age group and gender of the customer can be identified, and it is not necessary for an individual to be identified. Consequently, in general, a security-purpose surveillance camera and system, and a marketing-purpose observation camera and system are independently installed. As a result of such a

configuration, a system for marketing is such that personal information (image information by which an individual can be identified) captured for security purposes is prevented from being leaked.

[0005] As a related technique, in Patent Document 1 is disclosed a system that, with respect to image data captured at a kindergarten and the like, protects the privacy of other kindergarten children in a case where a guardian of a certain kindergarten child wishes to browse that image data by distributing image data in which masking has been performed with respect to the faces of the kindergarten children other than the child of that guardian.

PRIOR ART DOCUMENTS

Patent Documents

[0006] [Patent Document 1] Japanese Unexamined Patent Application, First Publication No. 2009-159221

SUMMARY OF INVENTION

Problem to be Solved by the Invention

[0007] As mentioned above, if a security camera and system for security and an observation camera and system for marketing are separately installed, there are problems such as high costs and installation space shortages since the required equipment costs, installation costs, operation costs, and installation space are doubled.

[0008] An exemplary object of the present invention is to provide an information output device, a camera, an information output system, an information output method, and a program that solve the problems mentioned above.

Means for Solving the Problem

[0009] An information output device according to a first exemplary aspect of the present invention includes: a first output unit that outputs acquired information acquired by a sensor; and a second output unit that converts personal information included in the acquired information into attribute information from which identification of an individual is impossible, and outputs the attribute information.

[0010] An information output device according to a second exemplary aspect of the present invention includes: an output unit that outputs at least one of acquired information acquired by a sensor, and attribute information obtained by converting personal information included in the acquired information into information from which identification of an individual is impossible.

[0011] A camera according to a third exemplary aspect of the present invention includes any one of the above-mentioned information output devices.

[0012] An information output system according to a fourth exemplary aspect of the present invention includes: an information output device including a first output unit and a second output unit, the first output unit encrypting acquired information acquired by a sensor and outputting the encrypted acquired information, the second output unit converting personal information included in the acquired information into attribute information from which identification of an individual is impossible and outputting the attribute information; a first server terminal device that acquires the acquired information output by the first output unit; and a second server terminal device that acquires the attribute information output by the second output unit. The first server terminal device includes a decryption unit that decrypts the acquired information.

[0013] An information output system according to a fifth exemplary aspect of the present invention includes: an information output device including an output unit that outputs at least one of acquired information acquired by a sensor, and attribute information obtained by converting personal information included in the acquired information into information from which identification of an individual is impossible; a first server terminal device that acquires the acquired information output by the output unit; and a second server terminal device that acquires the attribute information output by the output unit. The first server terminal device includes a decryption unit that decrypts the acquired information.

[0014] An information output method according to a sixth exemplary aspect of the present invention includes: outputting acquired information acquired by a sensor; and converting personal information included in the acquired information into attribute information from which identification of an individual is impossible, and outputting the attribute information.

[0015] An information output method according to a seventh exemplary aspect of the present invention includes: outputting at least one of acquired information acquired by a sensor, and attribute information obtained by converting personal information included in the acquired information into information from which identification of an individual is impossible.

[0016] A program according to an eighth exemplary aspect of the present invention causes a computer of an information output device to execute: outputting acquired information acquired by a sensor; and converting personal information included in the acquired information into attribute information from which identification of an individual is impossible, and outputting the attribute information.

[0017] A program according to a ninth exemplary aspect of the present invention causes a computer of an information output device to execute: outputting at least one of acquired information acquired by a sensor, and attribute information obtained by converting personal information included in the acquired information into information from which identification of an individual is impossible.

Effect of the Invention

[0018] According to an exemplary embodiment of the present invention, it is possible to output encrypted information of image data captured by a single camera and information in which personal information is converted into attribute information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a diagram showing a configuration of an edge terminal device according to a first exemplary embodiment of the present invention.

[0020] FIG. 2 is a diagram showing a detailed configuration of an edge terminal device according to the first exemplary embodiment of the present invention.

[0021] FIG. 3 is a diagram showing an example of an information output system according to the first exemplary embodiment of the present invention.

[0022] FIG. 4 is a diagram showing a process flow of the edge terminal device according to the first exemplary embodiment of the present invention.

[0023] FIG. 5 is a diagram showing a process flow of a server terminal device in which a security system according to the first exemplary embodiment of the present invention is in operation.

[0024] FIG. 6 is a diagram showing a process flow of a server terminal device in which a marketing system according to the first exemplary embodiment of the present invention is in operation.

[0025] FIG. 7 is a diagram showing an example of an information output system according to a second exemplary embodiment of the present invention.

[0026] FIG. 8 is a diagram showing an example of an information output system according to a third exemplary embodiment of the present invention.

[0027] FIG. 9 is a diagram showing a configuration of an edge terminal device according to a fifth exemplary embodiment of the present invention.

[0028] FIG. 10 is a diagram showing a configuration of a camera according to a sixth exemplary embodiment of the present invention.

EMBODIMENTS FOR CARRYING OUT THE INVENTION

First Exemplary Embodiment

[0029] Hereunder, an edge terminal device 1 according to a first exemplary embodiment of the

present invention is described with reference to FIG. 1 to FIG. 6.

[0030] FIG. 1 is a diagram showing a configuration of the edge terminal device 1 according to the first exemplary embodiment of the present invention.

[0031] As shown in FIG. 1, the edge terminal device 1 includes at least a first output unit 12 and a second output unit 13.

[0032] The first output unit 12 outputs acquired information that is acquired by a sensor.

[0033] The second output unit 13 converts personal information included in the acquired information that is acquired by the sensor into attribute information by which an individual cannot be identified, and then outputs it. The acquired information that is acquired by the sensor represents, for example, image data captured by a camera, audio collected by a microphone, a fingerprint, a palm print, a vein pattern, and an iris pattern.

[0034] FIG. 2 is a diagram showing a detailed configuration of an edge terminal device according to the first exemplary embodiment of the present invention.

[0035] As shown in FIG. 2, the edge terminal device 1 includes an information acquisition unit 11, the first output unit 12, the second output unit 13, a communication unit 14, an output determination unit 15, a storage unit 16, and a first connection unit 18.

[0036] The first connection unit 18 is connected to a sensor S installed in a space in which a customer is present.

[0037] The information acquisition unit 11 acquires from the sensor S, via the first connection unit 18, various sensing data (real-world data) related to the behavior of a customer and the environment surrounding the customer. The sensing data represents image data in which the customer has been captured for example. Hereunder, a case where the sensor S represents a camera, and the acquired information represents image data captured by the camera is described as an example. The image data includes video and still images.

[0038] The first output unit 12, for example, performs scrambling processing (encryption) with respect to image data acquired by the information acquisition unit 11, and outputs the data following scrambling processing to another device via the communication unit 14. The another device represents an information processing device such as a server that configures a security system of a security company for example.

[0039] The second output unit 13 performs personal information conversion processing, which converts personal information (information by which an individual can be identified) included in the acquired information into attribute information by which an individual cannot be identified. For example, the second output unit 13 may change a face of an individual appearing in the image data acquired by the information acquisition unit 11 into a mosaic, or replace it with an avatar. The second output unit 13 may analyze the image of a customer appearing in the image data using an image recognition technique to estimate the age group, gender, position information, and the like, of that customer. The second output unit 13 may calculate from the image data captured over a predetermined period, a flow line and at-shelf behavior of a customer appearing in that image data, and generate data (such as data converted to text) that represents the behavior of the customer as necessary for marketing. The second output unit 13 outputs the image data, in which personal information conversion processing has been performed, to another device via the communication unit 14. The another device represents, for example, an information processing device such as a server that configures a marketing system, which analyzes marketing information in a data center, and the like. A flow line represents information indicating a movement path of a customer. At-shelf behavior represents behavior relating to the selection of products performed by a customer in front of a product shelf.

[0040] The communication unit 14 performs communication with another device.

[0041] The output determination unit 15 determines whether or not data obtained by the personal information conversion processing performed by the second output unit 13, is to be output from the second output unit 13.

[0042] The storage unit **16** stores a program and various information.

[0043] The first output unit **12**, the second output unit **13**, and the output determination unit **15** are functions provided by a CPU (Central Processing Unit) provided in the edge terminal device **1** by executing a program.

[0044] FIG. **3** is a diagram showing an example of an information output system according to the first exemplary embodiment of the present invention.

[0045] The information output system shown in FIG. **3** includes at least one edge terminal device **1**, at least one camera (sensor) **4**, a server terminal device **2**, and a server terminal device **3**. The edge terminal device **1** is installed in a store **100**. The camera **4** is connected to the edge terminal device **1**. The server terminal device **2** is provided in a data center **200**. The server terminal device **3** is provided in a data center **300**. The edge terminal device **1** is installed comparatively near a customer, and performs processing that is appropriate to perform prior to transmission to the server terminal devices **2** and **3** of the data centers **200** and **300**. In the first exemplary embodiment, for example, the edge terminal device **1** performs processing that encrypts personal information, and performs processing that converts personal information into attribute information by which an individual cannot be identified. The edge terminal device **1**, the server terminal device **2**, and the server terminal device **3** are connected via a network NW. In the server terminal device **2**, a security system for surveillance is in operation. In the server terminal device **3**, a marketing system is in operation.

[0046] As shown in FIG. **3**, the server terminal device **2** includes a communication unit **21**, a decryption unit **22**, an application execution unit **23**, and a storage unit **24**. The communication unit **21** performs communication with another device, such as the edge terminal device **1** and the server terminal device **3**. The decryption unit **22** decrypts image data following scrambling processing (encrypted) that is acquired from the edge terminal device **1**. The application execution unit **23** executes respective application programs of the security system. The storage unit **24** stores the various application programs and decrypted image data.

[0047] As shown in FIG. **3**, the server terminal device **3** includes a communication unit **31**, an application execution unit **33**, and a storage unit **34**. The communication unit **31** performs communication with another device, such as the edge terminal device **1** and the server terminal device **2**. The application execution unit **33** executes respective application programs of the marketing system. The storage unit **34** stores the various application programs, and image data following personal information conversion processing that is acquired from the edge terminal device **1**.

[0048] The camera **4**, for example, is installed at a position overlooking the store **100**, and continuously captures a view of the interior of the store **100**. The image data captured by the camera **4** is used with an object of monitoring shoplifting, robberies, and the like. In addition, it is also used to grasp the behavior of customers that shop at the store **100**, with an object of utilization in marketing. The camera **4** outputs the captured image data to the edge terminal device **1**. The edge terminal device **1** performs scrambling processing for security with respect to the acquired image data. The edge terminal device **1** transmits the image data, in which scrambling processing has been performed, to the server terminal device (external to the edge terminal device **1**) **2**. Furthermore, the edge terminal device **1** performs personal information conversion processing for marketing with respect to the acquired image data. The edge terminal device **1** transmits the image data in which personal information conversion processing has been performed, to the server terminal device (external to the edge terminal device **1**) **3**.

[0049] In the server terminal device **2**, the security system is in operation. For example, a security guard monitors image data received by the server terminal device **2**. The image data received by the server terminal device **2** is stored for a predetermined period as evidence in a storage device provided in the data center **200** (or the storage unit **24**).

[0050] The server terminal device **3** analyzes, as objects, image data received from the edge

terminal device **1** and data in which image recognition has been performed and then converted to text. The server terminal device **3** may, based on the analysis result, deduce a top-selling product for each age group and gender. The server terminal device **3** may calculate, based on the analysis result, a location through which many customers pass, from the flow lines of the customers within the store, and perform marketing analysis that generates information that serves as a reference for product arrangement locations. A flow line represents information showing the path moved by a customer.

[0051] The security image data is preferably image data that includes personal information (an image by which a face can be recognized for example). This is because it is necessary for the security system to identify the person that carried out the shoplifting, and the like. However, if image data that includes personal information is transmitted as is from the edge terminal device **1** of the store **100** to the server terminal device **2** of the data center **200** via the network NW, for example, it is possible for that personal information to be leaked by means of a packet being captured during network transmission. Consequently, in the edge terminal device **1**, the first output unit **12** performs scrambling processing (encryption) with respect to the image data to be transmitted, by means of a predetermined method. The communication unit **14** transmits the image data following scrambling processing to the server terminal device **2**. As a result, it is possible to prevent personal information from being leaked during network transmission. The server terminal device **2** performs descrambling processing with respect to the received image data, to return it to the original image prior to scrambling processing. A decryption key that decrypts the image data in which scrambling processing was performed by the first output unit **12**, is held only by the server terminal device **2**. Consequently, only the server terminal device **2** is able to decrypt the acquired image data, and restore the original image data that includes personal information. On the other hand, even if a third party obtains the image data transmitted by the first output unit **12**, it is not possible to restore it to the original image data that includes personal information. Therefore, it is possible to prevent personal information from being leaked. The server terminal device **2** may enable monitoring by a security guard by utilizing the decrypted image data. The server terminal device **2** may also compare against image data of a person that has caused some problem behavior in the past who is registered in a so-called blacklist, and check if the person registered in the blacklist appears in the decrypted security image data.

[0052] Next, it is not necessary for the marketing data (or image data) to include data by which an individual can be identified. This is because in the marketing system it is sufficient if attribute information, such as the age group and gender of the customer, can be acquired. Furthermore, the transmission and reception of image data that includes personal information as is via the network NW, and storage in the server terminal device **3**, and the like, is a problem from the perspective of personal information being leaked. Therefore, the second output unit **13** extracts, with respect to the image data captured by the camera **4**, attribute information of a customer appearing in the image, which is configured by the image data, and transmits the extracted attribute information and the image data, which has been converted such that personal information cannot be recognized, to the server terminal device **3** of the data center **300**. The processing that performs conversion such that personal information cannot be recognized is performed by the second output unit **13**. The second output unit **13** may perform image processing that recognizes the face of a customer by means of image recognition, and then changes the area of the face into a mosaic for example. In this case, the second output unit **13** records the attribute information, such as the age group and gender of that customer, in the storage unit **16** such that it is associated with the identifying information of the same customer. The second output unit **13** may replace the face image of the customer with an avatar image. In this case, the avatar image that is substituted represents an avatar image that is associated with the attribute information such as the age group and the gender of that customer. The second output unit **13** may record the attribute information of the customer in the storage unit **16** such that it is associated with the identifying information of the avatar image. The

second output unit **13** transmits to the server terminal device **3**, together with the image data in which the conversion of personal information has been performed, data in which the attribute information of the customer has been associated with the customer or an avatar image. If the image data is such that the conversion of personal information has been performed, then there is no concern for leakage of personal information to occur even if it is stolen and accessed at the time of transmission and reception via the network NW. Furthermore, even if the image data, in which conversion of personal information has been performed, is stored on the server terminal device **3** for marketing analysis, there is no need for concern regarding the handling of personal information. [0053] Since, the server terminal device **3** acquires image data which has been converted such that personal information cannot be recognized, and attribute information of the customer, it can perform marketing analysis. For example, the server terminal device **3** may acquire image data following personal information conversion processing and the attribute information of the customer, and perform analysis of flow lines showing the movement path of the customer in the store, and of the at-shelf behavior, such as which products were picked up and then actually purchased or returned to the shelf. In the present specification, the attribute information not only represents the age group and gender of the customer, but also includes flow lines, the at-shelf behavior, the position information of the customer, and the like. That is to say, the second output unit **13** is provided with a function of performing marketing analysis, and it may perform, by means of that function, analysis of flow lines and at-shelf behavior and generate information including the age group, gender, flow lines, and at-shelf behavior of the customer as text data. Further, the second output unit **13** may transmit the text data to the server terminal device **3**.

[0054] Next, output processing by the edge terminal device **1** is described.

[0055] FIG. **4** is a first diagram showing a process flow of the edge terminal device **1** according to the first exemplary embodiment of the present invention.

[0056] The processing by which the edge terminal device **1** converts and outputs image data is described with reference to FIG. **4**. As a prerequisite, a state is assumed in which a view of the store **100** is being captured by the camera **4**.

[0057] Firstly, in the edge terminal device **1**, the information acquisition unit **11** acquires from the camera **4**, image data captured by the camera **4** (step S11). The information acquisition unit **11** outputs the acquired image data to the first output unit **12**. Then, the first output unit **12**, for example, acquires an encryption key recorded beforehand in the storage unit **16** (step S12). Next, the first output unit **12** uses the encryption key to perform scrambling processing with respect to the image data acquired from the information acquisition unit **11** (step S13). The encryption key is, for example, a public key or a common key provided by the security system-side. The scrambling processing may be performed using a well-known image encryption method. Then, the first output unit **12** transmits the security image data following scrambling processing, to the server terminal device **2** via the communication unit **14** (step S14).

[0058] The edge terminal device **1** performs the following processing concurrently with the processing of step S12 to step S14. Firstly, the output determination unit **15** determines whether or not to perform output of marketing image data from the second output unit **13** (step S15). For example, if the current time is not within the business hours of the store **100**, then there are no customers within the store **100** and purchasing behavior is not generated. Consequently, the utility value as image data for marketing is low. Therefore, in a time period where there are no customers, output of marketing image data is not necessarily performed. In this case, for example, the business hours of the store **100** are recorded beforehand in the storage unit **16**. The output determination unit **15** reads out information regarding the business hours from the storage unit **16**, and determines whether or not the current time is included in the business hours. If the current time is included in the business hours, the output determination unit **15** determines that output of marketing image data is to be performed. Even if the current time is within business hours, output of marketing image data may be not performed if there are no customers. In this case, a face image of the store clerk is

recorded beforehand in the storage unit **16**. The output determination unit **15** acquires image data from the image acquisition unit **11**, and confirms whether a person other than the store clerk appears in the acquired image data by means of image recognition. If a person other than the store clerk appears in the image data, the output determination unit **15** may determine that output of marketing image data is to be performed.

[0059] If the output determination unit **15** determines that output of marketing image data is to be performed (step **S15**: Yes), the information acquisition unit **11** outputs the acquired image data to the second output unit **13**.

[0060] The second output unit **13** performs personal information conversion processing with respect to the acquired image data (step **S16**). The second output unit **13** recognizes from the acquired image data, the people appearing in the picture, and estimates the age group and the gender of each person based on the face image of each person using a well-known face recognition technique or the like.

[0061] For example, the second output unit **13**, recognizes the color of the clothing worn by that person or the like by a well-known color recognition technique, and may, using this as a symbol to identify that person, store attribute information, such as the estimated gender, and the person, in the storage unit **16** such that they are associated. The information in which the attribute information and the person are associated represents, for example, information in which a person, such as “a customer wearing red clothing”, is associated with attribute information, such as “gender: female” and “age group: twenties”.

[0062] Alternatively, the second output unit **13** may perform the following processing. That is to say, the second output unit **13** reads out an avatar image stored beforehand that corresponds to the estimated age group and gender from the storage unit **16**. The second output unit **13** recognizes the face image of the person by means of a facial recognition technique. The second output unit **13** replaces the face image of the person with an avatar image that is similar to the contours of the face of that person.

[0063] As another alternative method, the second output unit **13** may perform the following processing in a case where it has a marketing analysis function. That is to say, the second output unit **13** acquires image data captured over a predetermined period. The second output unit **13** uses the clothing or the like worn by each person as a symbol to calculate position information of the person in each picture by, for example, a well-known technique such as a characteristic point extraction and object tracking technique. Further, the second output unit **13** may calculate a flow line for each person based on time series position information. The second output unit **13** may detect arm movements and face direction changes from a plurality of time series image data, and analyze the at-shelf behavior of the person by a movement in which the person extends a hand to the product shelf, by the direction in which the line-of-sight is directed, and the like. The second output unit **13** may generate data, in addition to the age group and the gender, in which information such as the flow line, and the at-shelf behavior, are converted to text. The data converted to text represents, for example, information such as “customer **001**, male, twenties, moved between shelf A and shelf B, picked up product C and purchased it”. The second output unit **13** outputs image data, which has been converted such that personal information cannot be recognized by the second output unit **13**, or data converted to text. The second output unit **13** transmits marketing image data (including attribute information and data converted to text) to the server terminal device **3** (step **S17**).

[0064] If the output determination unit **15** determines that output of marketing image data is to not be performed (step **S15**: No), the processing proceeds to step **S18**.

[0065] Then, for example, if the camera **4** ends imaging in accordance with an imaging stopping operation by an administrator and the like (step **S18**: Yes), the current process flow ends. If the camera **4** is in a period in which imaging is continued (step **S18**: No), the processing from step **S11** is repeated.

[0066] A case where, in step S13, an encryption key is recorded beforehand in the storage unit **16** was described as an example. However, the first exemplary embodiment is in no way limited to such a configuration. As an example, a case where the server terminal device **2** includes an encryption key generation unit **25** (refer to FIG. **3**) is described. In this case, the encryption key generation unit **25** generates an encryption key (one-time password) at predetermined intervals. The communication unit **21** transmits that encryption key. The communication unit **14** acquires that encryption key via the network NW. The first output unit **12** performs scrambling processing with that encryption key.

[0067] The output determination unit **15** may determine by means of image recognition whether or not a person is included in the image data, and the first output unit **12** be made to perform scrambling processing only in a case where a person is included in the image data. The first output unit **12** may recognize the face of a person by means of image recognition, and perform scrambling processing with respect to only the face of the person.

[0068] FIG. **5** is a diagram showing a process flow of a server terminal device **2** in which a security system according to the first exemplary embodiment of the present invention is in operation.

[0069] The processing up until the security system operated by the server terminal device **2** utilizes the image data acquired from the edge terminal device **1**, is described with reference to FIG. **5**.

[0070] Firstly, in the server terminal device **2**, the communication unit **21** acquires security image data via a network (step S21). The communication unit **21** outputs the acquired security image data to the decryption unit **22**. Then, the decryption unit **22**, for example, uses a private key corresponding to the public key used in the scrambling processing by the first output unit **12**, to perform descrambling processing that decrypts the acquired security image data (step S22). The decryption unit **22** records the image data, including the decrypted personal information, in the storage unit **24**. Next, the application execution unit **23** executes an application program of the security system, and utilizes the image data following descrambling (step S23). A specific example relating to the utilization of image data is described. For example, the application execution unit **23** may compare by means of a facial recognition technique, the image data of the person appearing in the image, which is configured by image data, against image data of a person registered in a blacklist and the like, to determine whether or not a person viewed as dangerous is present in the store **100**. As another example, in a case where a security guard performs real-time surveillance and discovers that a robbery, shoplifting, and the like, has occurred in the store **100**, the image data may be used to identify a person involved in the incident.

[0071] FIG. **6** is a diagram showing a process flow of a server terminal device **3** in which a marketing system according to the first exemplary embodiment of the present invention is in operation.

[0072] The processing up until the marketing system operated by the server terminal device **3** utilizes the image data acquired from the edge terminal device **1**, is described with reference to FIG. **6**.

[0073] Firstly, in the server terminal device **3**, the communication unit **31** acquires marketing image data via the network NW (step S31). The communication unit **31** records the acquired marketing image data in the storage unit **34**. Then, the application execution unit **33** executes an application program of the marketing system and utilizes the image data in which the personal information has been converted to attribute information, and the attribute information (step S32). A specific example of utilization of image data and attribute information is described. For example, the application executing unit **33** may generate information that is useful for promoting sales by predicting product demand from the attribute information (age group, gender, flow line, at-shelf behavior) of the customer appearing in the image data, and analyzing purchasing trends according to the age group, the gender, the region, the season, and the like.

[0074] According to the first exemplary embodiment, it is possible to output data in which personal information for security is included, and data in which attribute information of a customer for

marketing is included, without separately installing a security camera and system for security, and an observation camera and system for marketing.

Second Exemplary Embodiment

[0075] Hereunder, an information output system according to a second exemplary embodiment of the present invention is described with reference to FIG. 7.

[0076] FIG. 7 is a diagram showing an example of the information output system according to the second exemplary embodiment of the present invention. The sections in which the second exemplary embodiment are different from the first exemplary embodiment are described with reference to FIG. 7.

[0077] As shown in FIG. 7, in the information output system according to the second exemplary embodiment, an external terminal unit 5 is connected to the edge terminal device 1. The external terminal unit 5 is, for example, provided from the administrator of the security system of the server terminal device 2 to the store 100 that utilizes the services of that security system. The external terminal unit 5 represents a USB dongle for example. The external terminal unit 5 has recorded in it, an encryption program that performs scrambling processing, and an encryption key. The first output unit 12 of the edge terminal device 1 calls and executes the encryption program recorded in the external terminal unit 5, and performs scrambling processing with respect to the image data acquired from the information acquisition unit 11.

[0078] In a case where the external terminal unit 5 is not connected, the first output unit 12 does not operate, and output of image data is unable to be performed. In the server terminal device 2, a decryption key corresponding to the encryption key, which is recorded in the external terminal unit 5, is recorded in the storage unit 24. The decryption unit 22 performs descrambling processing with respect to the image data using the decryption key.

[0079] In this manner, a portion of the scrambling processing function of the first output unit 12 delegated to the external terminal unit 5. Further, the provider of the external terminal unit 5 is the administrator of the security system. Consequently, the administrator of the security system is able to conceal information related to the encryption of the image data. As a result, security is more easily assured. Furthermore, it is not necessary for the edge terminal device 1 to implement a scrambling processing function. Therefore, development costs can be inhibited.

Third Exemplary Embodiment

[0080] Hereunder, an information output system of a third exemplary embodiment of the present invention is described with reference to FIG. 8.

[0081] FIG. 8 is a diagram showing an example of the information output system according to the third exemplary embodiment of the present invention. The sections in which the third exemplary embodiment are different from the first and second exemplary embodiments are described with reference to FIG. 8.

[0082] As shown in FIG. 8, in the information output system according to the third exemplary embodiment, a card reading device 6 is connected to the edge terminal device 1. That is to say, the edge terminal device 1 has a second connection unit 19 to which the card reading device 6 is connected. The card reading device 6 is equipped with a key card 7 in which an encryption key is recorded. The first output unit 12 of the edge terminal device 1 reads out the encryption key recorded in the key card 7 via the second connection unit 19. The first output unit 12 utilizes the encryption key that was read out, to perform scrambling processing with respect to image data acquired from the information acquisition unit 11. In the same manner, a card reading device 8 is connected to the server terminal device 2. That is to say, the server terminal device 2 has a connection unit 26 to which the card reading device 8 is connected. The card reading device 8 is equipped with a key card 9 for example. A decryption key corresponding to the encryption key of the key card 7 is recorded in the key card 9. In the server terminal device 2, the decryption unit 22 uses the decryption key recorded in the key card 9, to decrypt the image data following scrambling processing acquired from the edge terminal device 1. An IC card may be used as the key card 7 and

the key card **9** for example.

[0083] In this manner, in the third exemplary embodiment, the encryption key is modularized and recorded in a detachable medium (key card). As a result of using key cards **7** and **9** in which a digital certificate issued by an official authentication organization is stored, the reliability of the encryption system is assured, and it is possible to reduce the risk of personal information being leaked.

Fourth Exemplary Embodiment

[0084] The first to third exemplary embodiments were described using an example of a case where the acquired information that is acquired by the information acquisition unit **11** is image data. The fourth exemplary embodiment is an exemplary embodiment in which the acquired information is not image data.

[When Acquired Information is Audio]

[0085] In a case where the acquired information is audio, the sensor **S** connected to the edge terminal device **1** represents, for example, a microphone provided with a sound collection function. The first output unit **12** encrypts the audio data that is acquired by the microphone, and then outputs it. The second output unit **13** estimates the age group and gender of the speaker from the acquired audio data, associates the estimated information with that audio data, and records it in the storage unit **16**. The second output unit **13** performs, as personal information conversion processing, processing such as band compression or noise addition with respect to the acquired audio data, such that an individual cannot be identified by voiceprint analysis. Then, the second output unit **13** outputs the audio data after personal information conversion processing has been performed. The second output unit **13** may output attribute information together with the audio data after the personal information conversion processing has been performed. The second output unit **13** may, in addition to the age group of the speaker, convert the spoken content to text data by means of a voice recognition technique, and then output the text data, and the like.

[0086] In a case where the acquired information is audio, information such as the gender, the age group, and the spoken content represent attribute information.

[0087] In the server terminal device **2**, the decryption unit **22** decrypts the encrypted audio. An application program of the security system that is executed by the application execution unit **23** may use the audio data following decryption to perform surveillance. The application execution unit **23** may record the audio data following decryption in the storage unit **24** as evidence.

[0088] In the server terminal device **3**, an application program of the marketing system executed by the application execution unit **33** may use the attribute information and the audio data following conversion, or the text data from voice recognition, acquired via the communication unit **31**, to analyze the needs of a customer that has visited the store **100**, and the like. For example, as a result of the store clerks wearing a microphone, and performing analysis of a conversation between a store clerk and a customer, it is possible to analyze the types of products the customer is searching for, or a complaint from the customer, and to use this to create better stores.

[When Acquired Information is Fingerprint, Palm Print, Vein Pattern, Iris Pattern, and the Like]

[0089] In a case where the acquired information is fingerprint data, the sensor **S** represents a fingerprint sensor. Currently, the use of biometric authentication by means of fingerprint data, and the like, is progressing. A specific example of a case where a fingerprint sensor is installed in a store **100** is described below.

[0090] In the store **100**, a special customer system which provides a variety of benefits is introduced. As a condition of a certain customer becoming a special customer of the store **100**, fingerprint data of that customer, and personal information of that customer are recorded beforehand in the storage unit **16** of the edge terminal device **1** such that they are associated. When that customer performs authentication by bringing a finger into contact with a fingerprint sensor at an entrance or at a register, points are added each time that customer comes to the store for example. If fingerprint authentication is performed at a register, points corresponding to the

purchase amount are applied. In this case, the first output unit **12** encrypts the fingerprint data acquired from the fingerprint sensor connected to the edge terminal device **1**, and then outputs it. From the acquired fingerprint data, the second output unit **13** acquires information such as the age group and gender of the customer from the personal information of the customer, which is stored in the storage unit **16** of the edge terminal device **1** in association with the fingerprint data. In the case where the storage unit **16** records for each customer, information such as whether that customer is a good customer, and the number of store visits, such that it is associated with the fingerprint data, the second output unit **13** may read out these pieces of information. Such information, that is to say, information such as the gender, the age group, whether or not the customer is a good customer, and the number of store visits, represents attribute information. The second output unit **13** outputs the attribute information corresponding to the fingerprint data acquired from the information acquisition unit **11**.

[0091] In the server terminal device **2**, the decryption unit **22** decrypts the encrypted fingerprint data. An application program of the security system executed by the application execution unit **23** records the fingerprint data following decryption in the storage unit **24**, for example, such that it is associated with the position information of the store **100** and the time in which the fingerprint data was acquired. Consequently, it becomes evidence that the customer was present in the store **100** at that time.

[0092] In the server terminal device **3**, an application program of the marketing system executed by the application execution unit **33** uses the attribute information acquired via the communication unit **31** to perform analysis of the purchasing behavior of the special customers visiting the store **100**, and the like. For example, if the store **100** is a chain store, it is possible to know the purchasing behavior of the special customers across a plurality of stores. A case where the acquired information represents a palm print, a vein pattern, or an iris pattern, is the same as the case of fingerprint data.

Fifth Exemplary Embodiment

[0093] A fifth exemplary embodiment is an exemplary embodiment of a case where there is only one output unit. Hereunder, an edge terminal device **1** according to the fifth exemplary embodiment of the present invention is described with reference to FIG. **9**.

[0094] FIG. **9** is a diagram showing a configuration of the edge terminal device **1** according to the fifth exemplary embodiment of the present invention.

[0095] As shown in FIG. **9**, the edge terminal device **1** includes at least an output unit **17**.

[0096] The output unit **17** outputs, at least one of first output information, which outputs acquired information that is acquired by a sensor, and second output information, in which personal information included in the acquired information that is acquired by the sensor is converted into attribute information by which an individual cannot be identified.

[0097] An example of a specific configuration of the edge terminal device **1** in the fifth exemplary embodiment is described. The edge terminal device **1** in the fifth exemplary embodiment, in addition to the output unit **17**, includes, an information acquisition unit **11**, a communication unit **14**, an output determination unit **15**, and a storage unit **16**. The configuration of these is the same as the configuration described with reference to FIG. **2**.

[0098] A process flow of the fifth exemplary embodiment is described with reference to FIG. **4**. Firstly, the information acquisition unit **11** acquires image data captured by the camera **4** (step **S11**). The information acquisition unit **11** outputs the acquired image data to the output unit **17**. Then, the output unit **17** acquires an encryption key (step **S12**), and performs scrambling processing with respect to the image data acquired from the information acquisition unit **11** (step **S13**). Next, the output unit **17** selects the communication interface of the communication unit **14** that is connected to the server terminal device **2**, and transmits security image data following scrambling processing (first output information) to the server terminal device **2** via that communication interface (step **S14**).

[0099] The output determination unit **15** determines whether or not to output marketing image data (step **S15**), and outputs the result thereof to the output unit **17**. The output determination unit **15** may determine, by means of image recognition, whether or not a person is included in the image data, and it may determine that marketing image data is to be output only in a case where a person is included in the image data. If the output determination unit **15** determines that marketing image data is to be output, the output unit **17** performs personal information conversion processing with respect to the image data acquired from the information acquisition unit **11** (step **S16**), and generates marketing image data (including attribute information and data converted to text). Then, the output unit **17** selects the communication interface of the communication unit **14** that is connected to the server terminal device **3**, and transmits the generated marketing image data (second output information) to the server terminal device **3** via that communication interface (step **S17**). The other processing steps are the same as in the first exemplary embodiment.

[0100] The output determination unit **15** may determine, by means of image recognition, whether or not a person is included in the image data, and it may transmit security image data to the server terminal device **2** only in a case where a person is included in the image data.

Sixth Exemplary Embodiment

[0101] Hereunder, a camera **5** according to a sixth exemplary embodiment of the present invention is described with reference to FIG. **10**.

[0102] As shown in FIG. **10**, the camera **5** includes an edge terminal device **1** and a camera function unit **51**. The configuration and the functions of the edge terminal device **1** shown in FIG. **10** are the same as those of the edge terminal device **1** shown in FIG. **1** except in the aspect that the camera function unit **51** is connected to the first connection unit **18** instead of the sensor **S**. The configuration and the functions of the camera function unit **51** are the same as the camera **4** shown in FIG. **3**.

[0103] The edge terminal device **1** mentioned above has a computer in its interior. Further, the processing steps of the edge terminal device **1** mentioned above are recorded in a computer-readable recording medium in the form of a program, and the processing described above is performed by means of the computer reading the program and then executing it. Here, the computer-readable recording medium refers to a magnetic disk, a magnetic optical disk, a CD-ROM, a DVD-ROM, a semiconductor memory, and the like. Furthermore, the computer program may be distributed to a computer by means of a communication line, and the computer receiving the distribution may execute that program.

[0104] The program described above may represent a program for realizing a portion of the functions mentioned above.

[0105] The program described above may be one that realizes the functions mentioned above by being combined with a program already stored in the computer system, as a so-called difference file (difference program).

[0106] In addition, the configuration elements in the exemplary embodiments described above may be appropriately substituted with well-known configuration elements without departing from a scope of the present invention. For example, the video data may be also audio data or image data, such as a still image. Furthermore, the technical scope of the present invention is in no way limited to the exemplary embodiments described above, and various changes may be applied without departing from the scope of the present invention. The edge terminal device **1** is an example of an information output device. The server terminal device **2** is an example of a first server terminal device. The server terminal device **3** is an example of a second server terminal device. The external terminal unit **5** is an example of an external connection device.

[0107] This application is based upon and claims the benefit of priority from Japanese patent application No. 2015-005538, filed Jan. 15, 2015, the disclosure of which is incorporated herein in its entirety by reference.

Industrial Applicability

[0108] The present invention may be applied to an information output device, a camera, an information output system, an information output method, and a program.

REFERENCE SYMBOLS

[0109] **1** Edge terminal device [0110] **2** Server terminal device [0111] **3** Server terminal device [0112] **4** Camera [0113] **5** External terminal unit [0114] **6, 8** Card reading device [0115] **7, 9** Key card [0116] **11** Information acquisition unit [0117] **12** First output unit [0118] **13** Second output unit [0119] **14** Communication unit [0120] **15** Output determination unit [0121] **16** Storage unit [0122] **18** First connection unit [0123] **19** Second connection unit [0124] **21, 31** Communication unit [0125] **22** Decryption unit [0126] **23, 33** Application execution unit [0127] **24, 34** Storage unit [0128] **26** Connection unit [0129] **51** Camera function unit

Claims

1. An information processing system comprising: a first terminal including at least one first memory storing first instructions and at least one first processor; a second terminal including at least one second memory storing second instructions and at least one second processor; and at least one camera including at least one third memory storing third instructions and at least one third processor, wherein the at least one third processor is configured to execute the third instructions to: acquire one or more images; encrypt the one or more images by using encryption key; analyze, based on the one or more images, attribute information; determine whether or not a person other than a store clerk appears in the acquired one or more images and whether or not a current time is within business hours of a store; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, do not output the encrypted one or more images and the attribute information in a case where the current time is not within the business hours; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, do not output the encrypted one or more images and the attribute information in a case where the current time is within the business hours and the person other than the store clerk is not included in the acquired one or more images; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, output the encrypted one or more images and the attribute information in a case where the current time is within the business hours and the person other than the store clerk is included in the acquired one or more images; perform, by using a decryption key corresponding to the encryption key, decrypting processing with respect to the encrypted one or more images; determine whether or not the person included in the decrypted one or more images is registered in a list by comparing the person included in the decrypted one or more images with the list, the list including a plurality of images of people; and output a determination result of whether or not the person is registered in the list, wherein the at least first processor is configured to execute the first instructions to acquire, from the at least one camera, first information by which a person can be identified, and the first information being used for surveillance, wherein the at least second processor is configured to execute the second instructions to acquire, from the at least one camera, second information used for marketing analysis, wherein the second information is generated by analyzing the attribute information based on a movement path of the person included in the one or more images, and wherein the attribute information is information acquired by converting personal information such that identity of the person is not revealed, the personal information being included in the one or more images.

2. The information processing system according to claim 1, wherein the at least one camera is provided in a store, wherein the at least one first processor is configured to execute the first instructions to: compare image data of the person with image data of other person registered in

advance; and determine whether or not the person is present at the store.

3. The information processing system according to claim 1, wherein the attribute information represents at least any one of flow lines of the person or at-shelf behavior of the person.

4. The information processing system according to claim 1, wherein the at least one camera comprises at least one third memory storing third instructions and at least one third processor configured to execute the third instructions to encrypt the one or more images by using an encryption key transmitted from the first terminal.

5. The information processing system according to claim 4, wherein the encryption key includes one-time password generated by the first terminal at predetermined intervals.

6. The information processing system according to claim 1, wherein encrypting the one or more images includes scrambling processing for the one or more images, wherein the at least one camera comprises at least one third memory storing third instructions and at least one third processor configured to execute the third instructions to perform scrambling processing with respect to a face image of the person included in the one or more images.

7. The information processing system according to claim 1, wherein the second output information is generated based on flow lines and at-shelf behavior of the person.

8. The information processing system according to claim 1, wherein the attribute information represents at least any one of an age group and gender.

9. The information processing system according to claim 1, wherein the at least one second processor is configured to execute the second instructions to output text data which represents a result of the marketing analysis.

10. The information processing system according to claim 1, wherein the attribute information is information based on a change in a position of the person.

11. An information processing method for controlling an information processing system including a first terminal, a second terminal, and at least one camera, the method comprising: acquiring one or more images by the at least one camera; encrypting the one or more images by using encryption key; analyzing, based on the one or more images, attribute information; determining whether or not a person other than a store clerk appears in the acquired one or more images and whether or not a current time is within business hours of a store; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, do not outputting the encrypted one or more images and the attribute information in a case where the current time is not within the business hours; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, do not outputting the encrypted one or more images and the attribute information in a case where the current time is within the business hours and the person other than the store clerk is not included in the acquired one or more images; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, outputting the encrypted one or more images and the attribute information in a case where the current time is within the business hours and the person other than the store clerk is included in the acquired one or more images; performing, by using a decryption key corresponding to the encryption key, decrypting processing with respect to the encrypted one or more images; determining whether or not the person included in the decrypted one or more images is registered in a list by comparing the person included in the decrypted one or more images with the list, the list including a plurality of images of people; and outputting a determination result of whether or not the person is registered in the list, acquiring, from the at least one camera, first information by which a person can be identified, and the first information being used for surveillance; and acquiring, from the at least one camera, second information used for marketing analysis, wherein the second information is generated by analyzing the attribute information based on a movement path of the person included in the one or more images, and

wherein the attribute information is information acquired by converting personal information such that identity of the person is not revealed, the personal information being included in the one or more images.

12. The information processing method according to claim 11, wherein the at least one camera is provided in a store, wherein the method further comprises: comparing image data of the person with image data of other person registered in advance; and determining whether or not the person is present at the store.

13. The information processing method according to claim 11, wherein the attribute information represents at least any one of flow lines of the person or at-shelf behavior of the person.

14. The information processing method according to claim 11, further comprising encrypting the one or more images by using an encryption key transmitted from the first terminal.

15. The information processing method according to claim 14, wherein the encryption key includes one-time password generated by the first terminal at predetermined intervals.

16. The information processing method according to claim 11, wherein encrypting the one or more images includes scrambling processing for the one or more images, wherein the method further comprises performing scrambling processing with respect to a face image of the person included in the one or more images.

17. The information processing method according to claim 11, wherein the second output information is generated based on flow lines and at-shelf behavior of the person.

18. The information processing method according to claim 11, wherein the attribute information represents at least any one of an age group and gender.

19. The information processing method according to claim 11, further comprising outputting text data which represents a result of the marketing analysis.

20. A non-transitory storage medium storing a program executing a method for controlling an information processing system including a first terminal, a second terminal, and at least one camera, the method comprising: acquiring one or more images by the at least one camera; encrypting the one or more images by using encryption key; analyzing, based on the one or more images, attribute information; determining whether or not a person other than a store clerk appears in the acquired one or more images and whether or not a current time is within business hours of a store; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, do not outputting the encrypted one or more images and the attribute information in a case where the current time is not within the business hours; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, do not outputting the encrypted one or more images and the attribute information in a case where the current time is within the business hours and the person other than the store clerk is not included in the acquired one or more images; based on the determination of whether or not the person other than the store clerk appears in the acquired one or more images and whether or not the current time is within the business hours of the store, outputting the encrypted one or more images and the attribute information in a case where the current time is within the business hours and the person other than the store clerk is included in the acquired one or more images; performing, by using a decryption key corresponding to the encryption key, decrypting processing with respect to the encrypted one or more images; determining whether or not the person included in the decrypted one or more images is registered in a list by comparing the person included in the decrypted one or more images with the list, the list including a plurality of images of people; and outputting a determination result of whether or not the person is registered in the list, acquiring, from the at least one camera, first information by which a person can be identified, and the first information being used for surveillance; and acquiring, from the at least one camera, second information used for marketing analysis, wherein the second information is generated by analyzing the attribute information based on a movement

path of the person included in the one or more images, and wherein the attribute information is information acquired by converting personal information such that identity of the person is not revealed, the personal information being included in the one or more images.
